$$Q \in \mathbb{R}^{n_q \times d_k}, \quad K \in \mathbb{R}^{n_k \times d_k},$$
$$V \in \mathbb{R}^{n_k \times d_v},$$
$$A = \mathrm{softmax}_{\mathrm{row}}\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_k}}\right),$$
$$\mathrm{Attention}(Q, K, V) = AV$$